

3 Nervous communication involves neurones and changes in ion concentrations.

(a) (i) Which description of the movement of ions will produce an action potential in an axon?

(1)

- ☐ **A** K^+ ions move into the axon
- ☐ **B** K^+ ions move out of the axon
- ☐ **C** Na^+ ions move into the axon
- ☐ **D** Na^+ ions move out of the axon

(ii) Which row describes the events that occur during repolarisation of a neurone?

(1)

	Sodium ion channel	Membrane potential
☐ A	closed	decreasing
☐ B	open	decreasing
☐ C	open	increasing
☐ D	closed	increasing

(b) Describe the structure of a spinal reflex arc.

(3)

(c) Explain how the nerve impulse is transmitted across a synapse.

(4)

(Total for Question 3 = 9 marks)